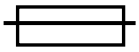
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

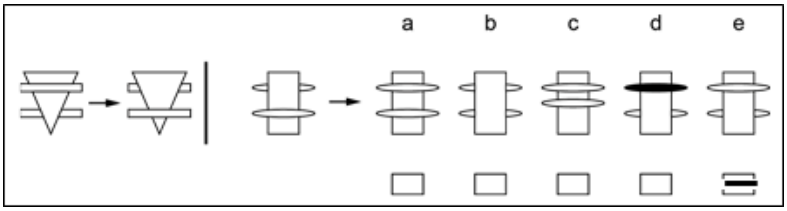
- You have 25 minutes to complete the paper.
- Mark your answer like this: 
- This paper features a total of 39 questions from 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't answer a question, move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

© Learning Street

1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

Example



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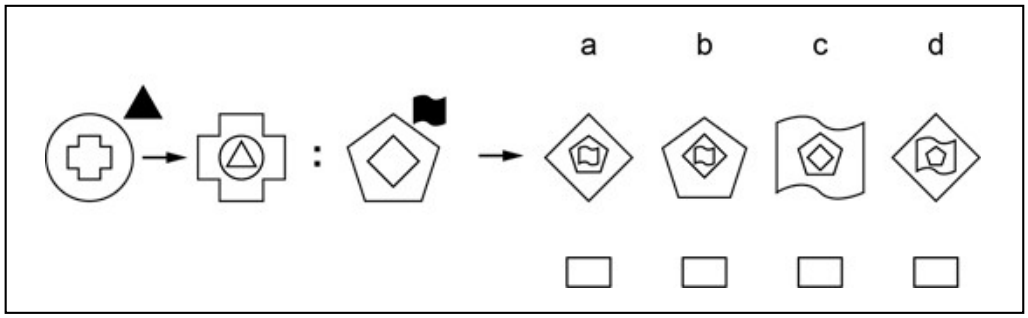

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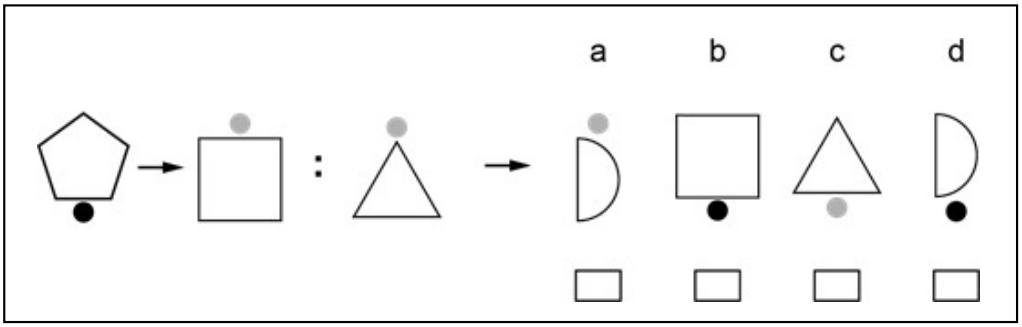
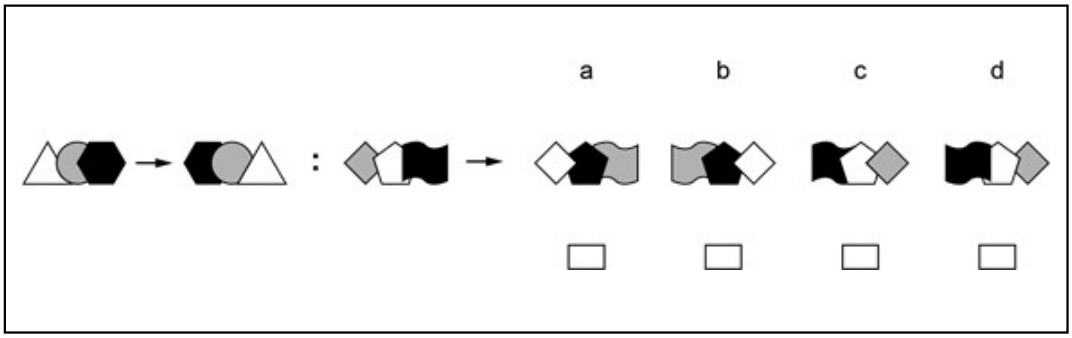


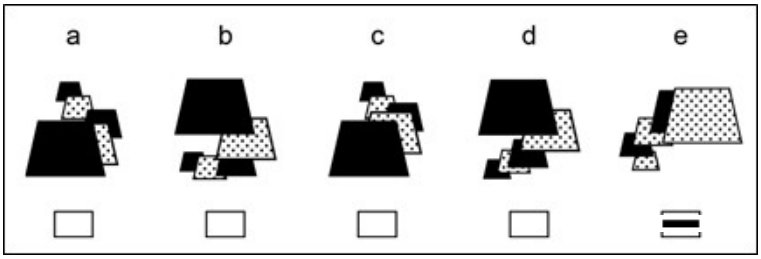
3



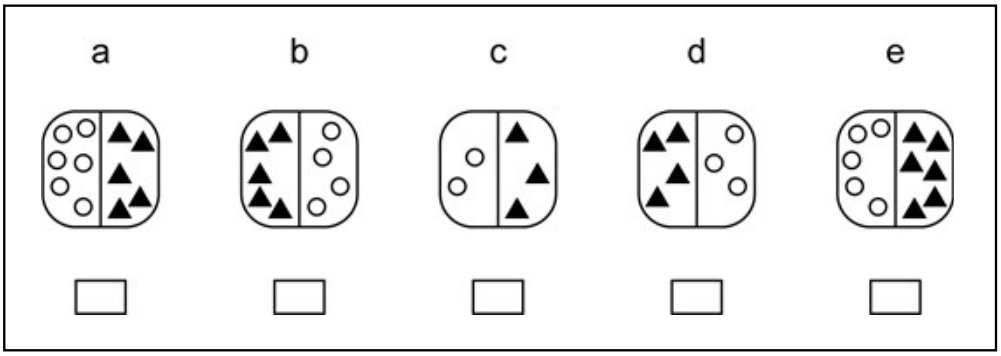
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

Example



1



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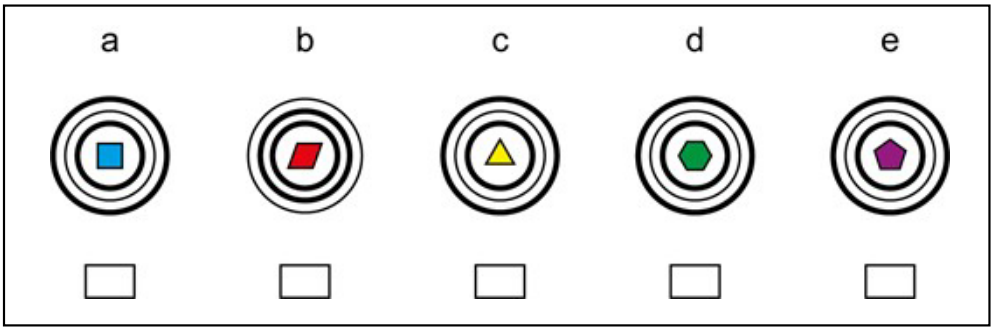
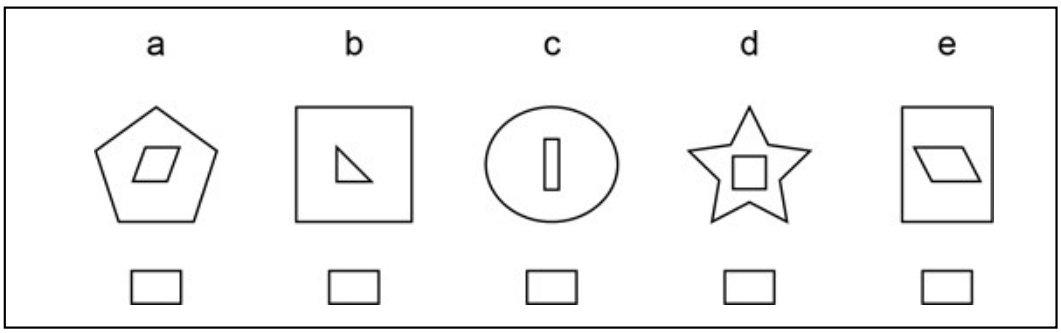


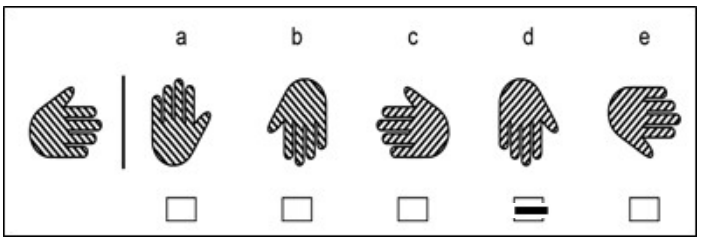
3



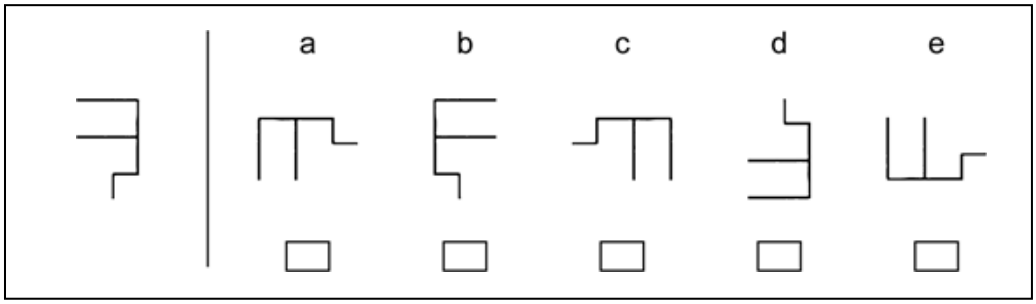
3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

Example



1



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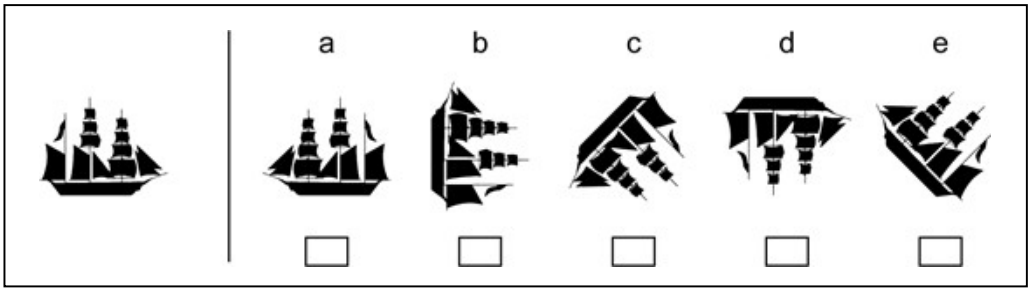
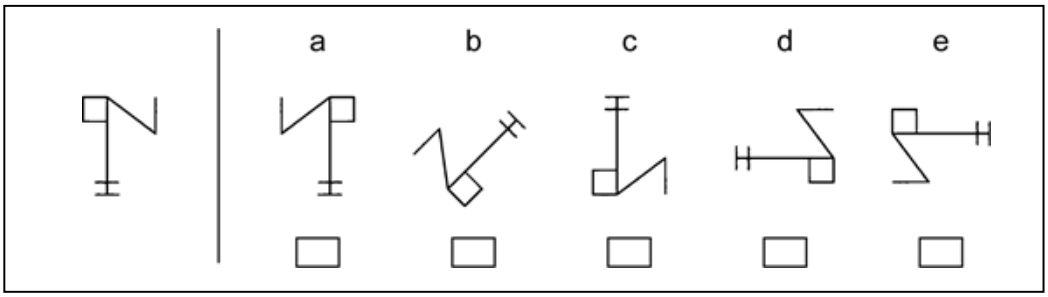


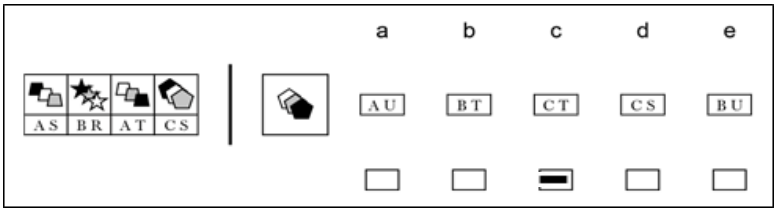
3



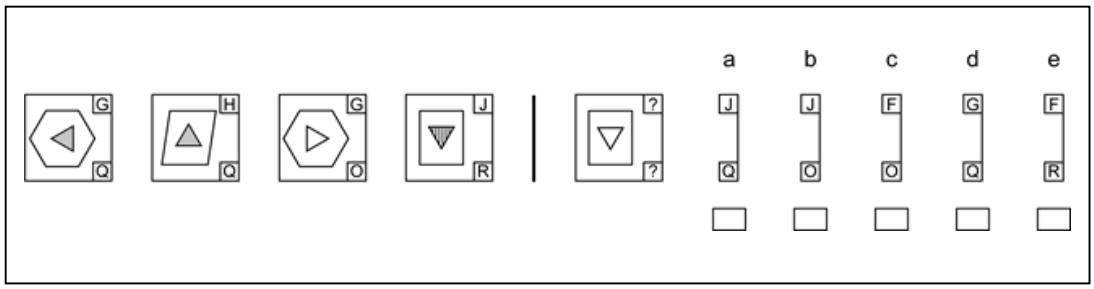
4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

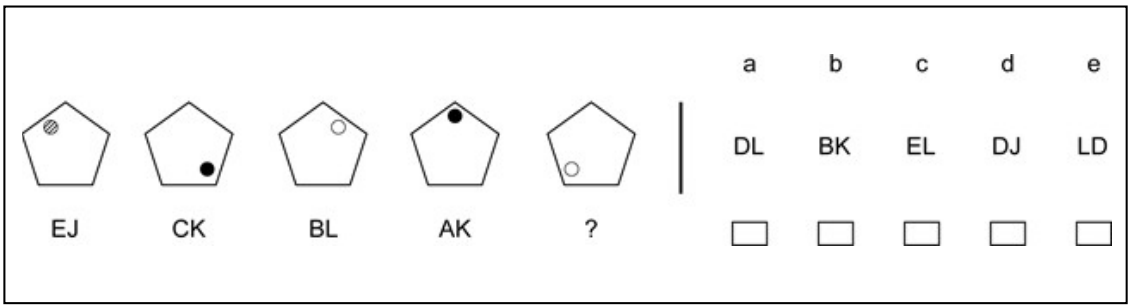
Example



1



2



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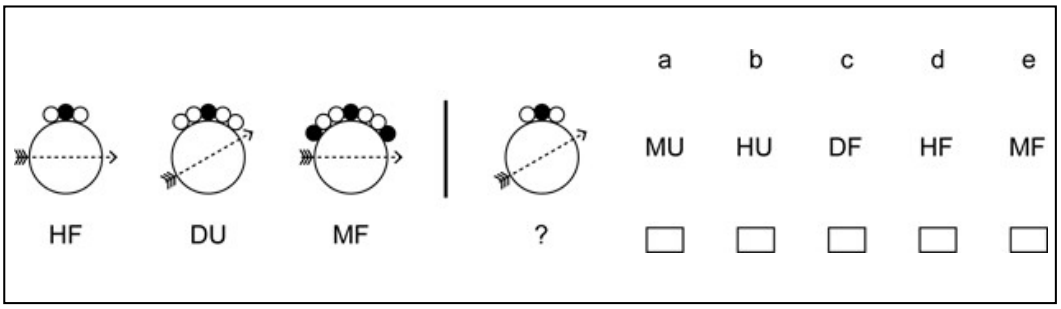
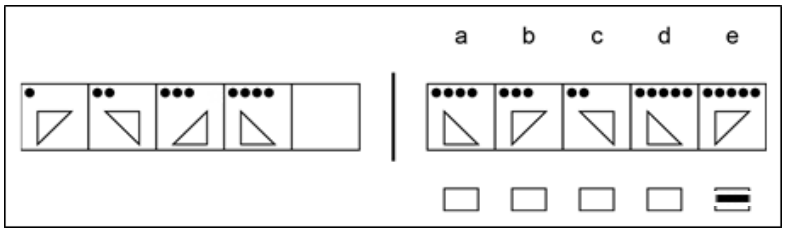


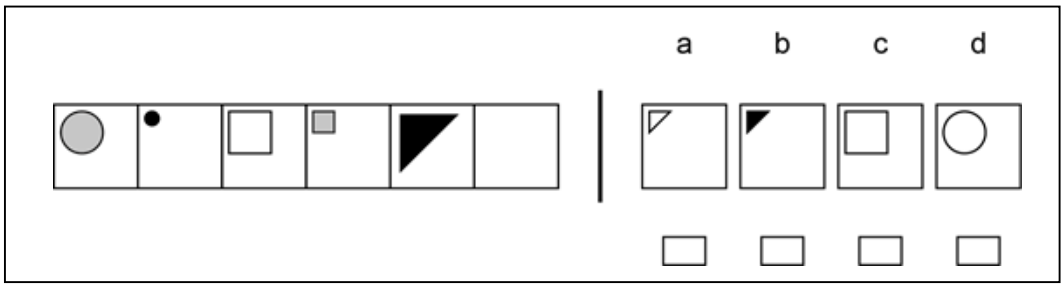
5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

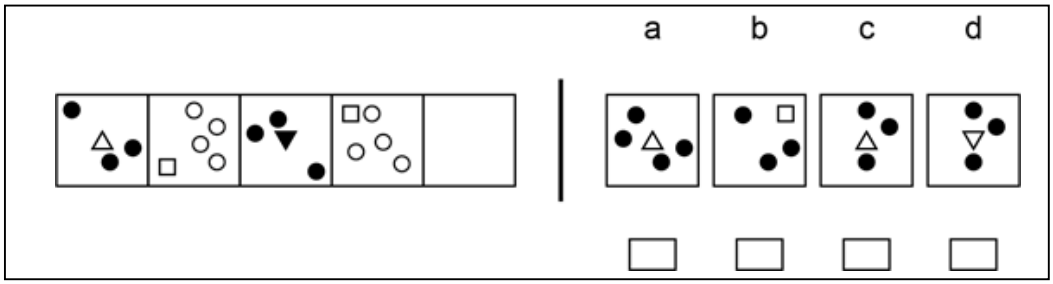
Example



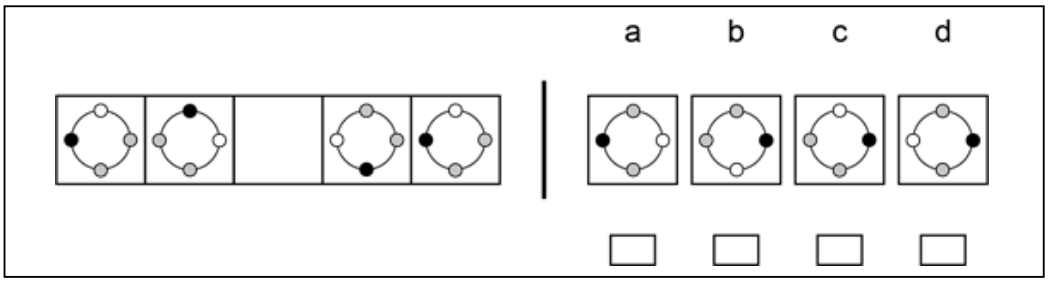
1



2



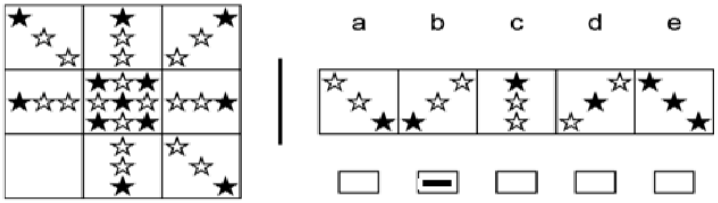
3



6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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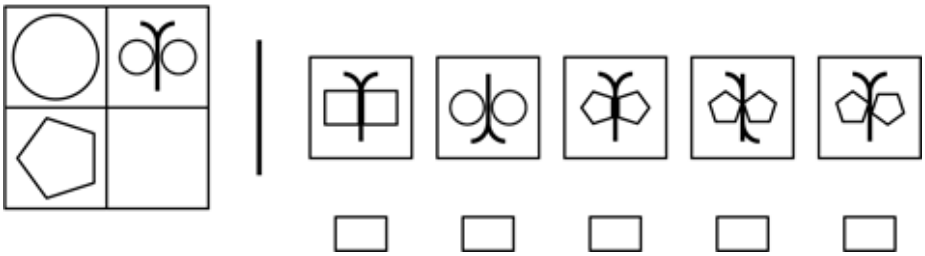
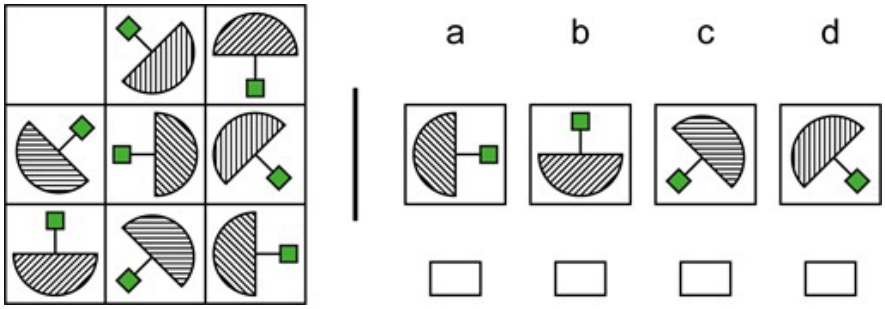


2

5		29
7	19	13
9	3	2

a	b	c	d	e
27	17	12	15	23

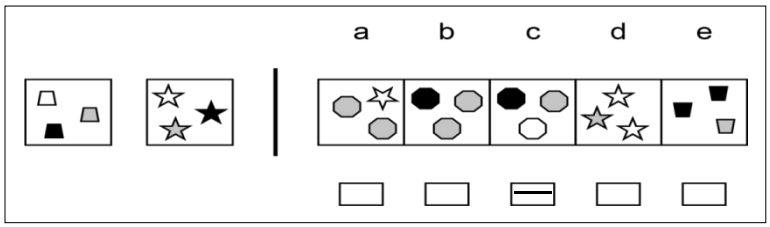
3



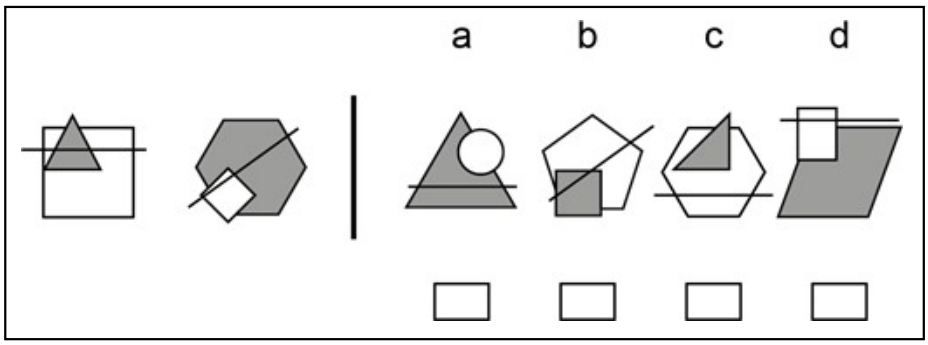
7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

Example



1



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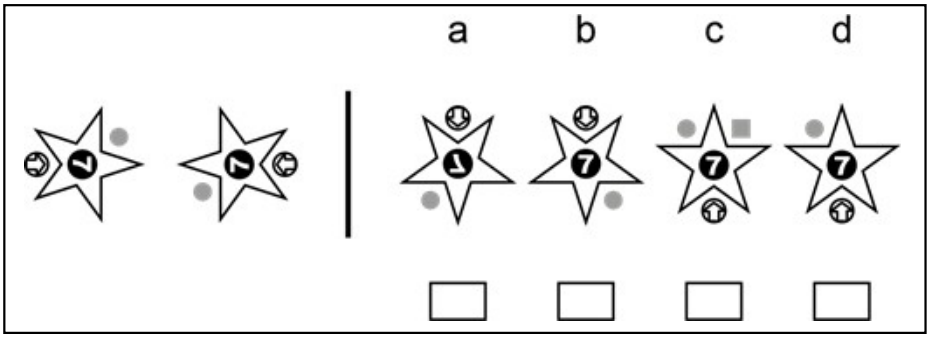
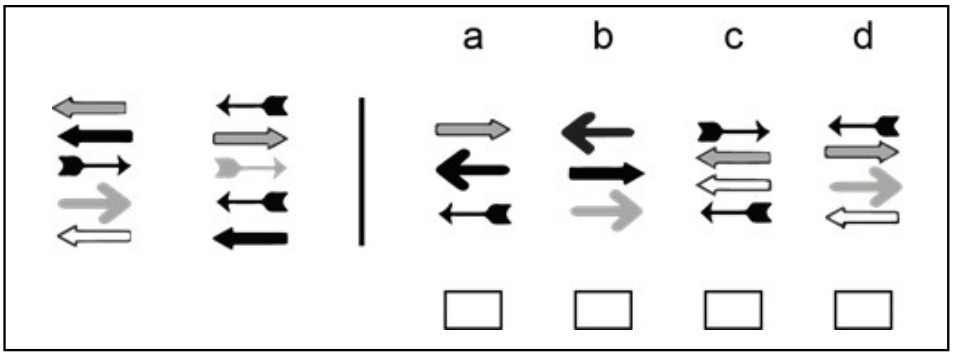


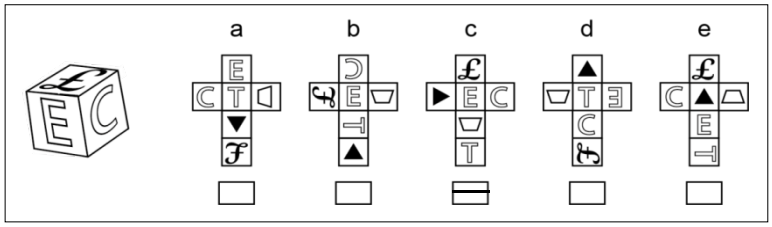
3



8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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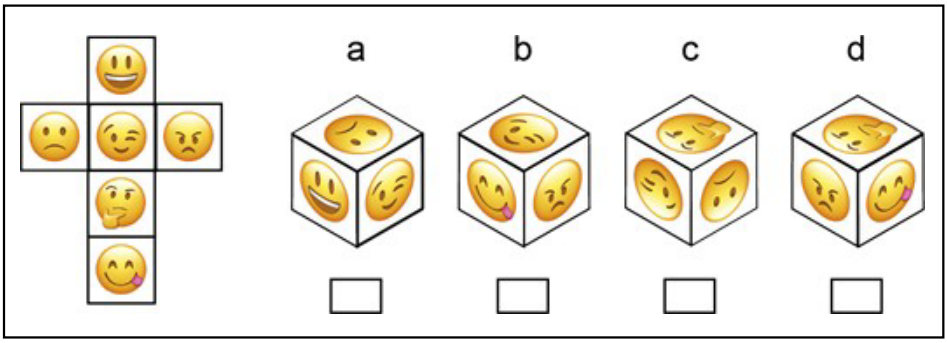
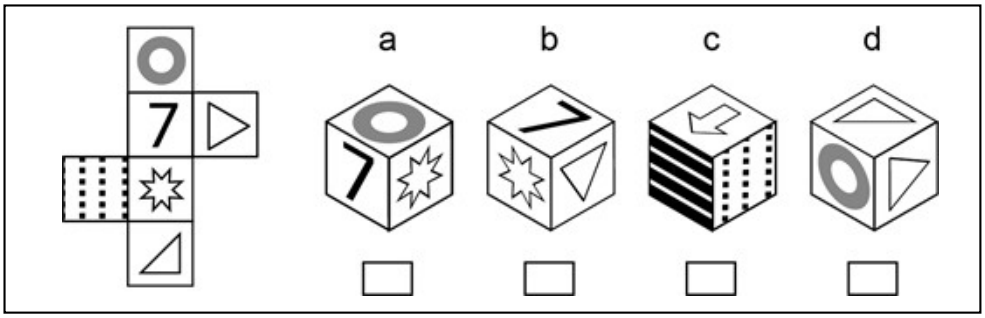
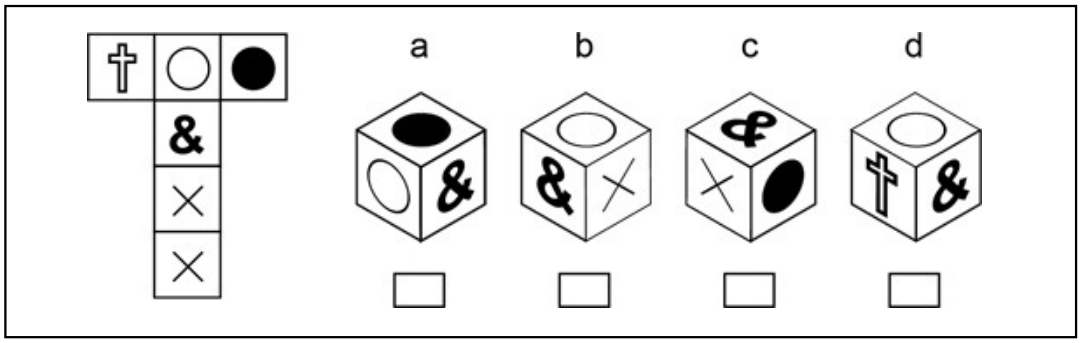


2



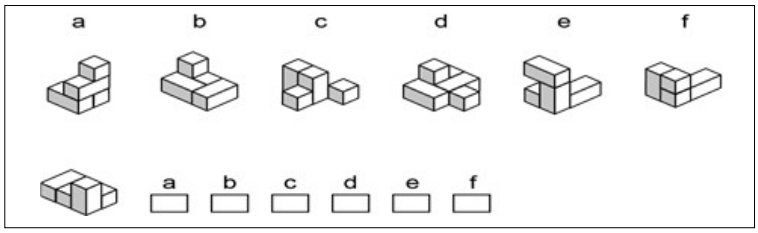
3



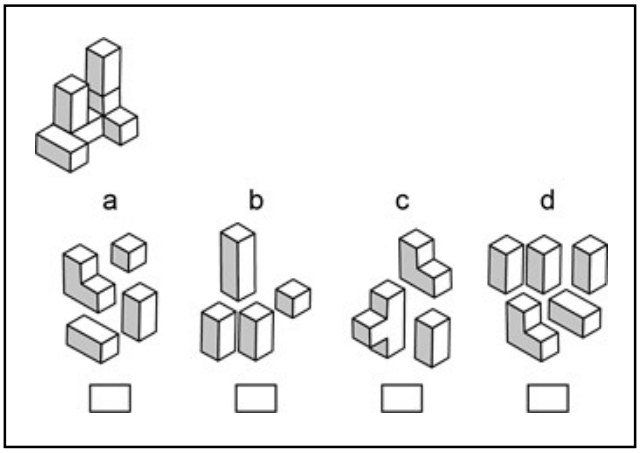
9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

Example



1



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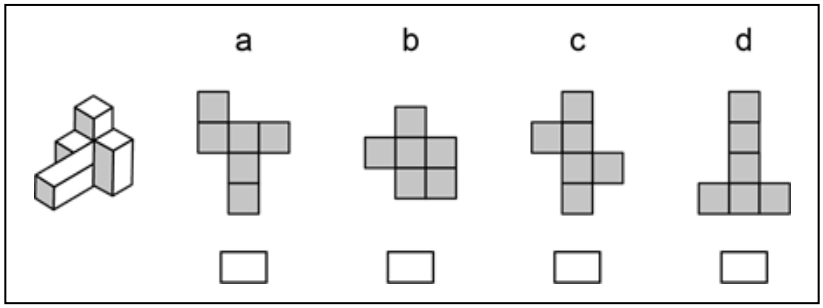

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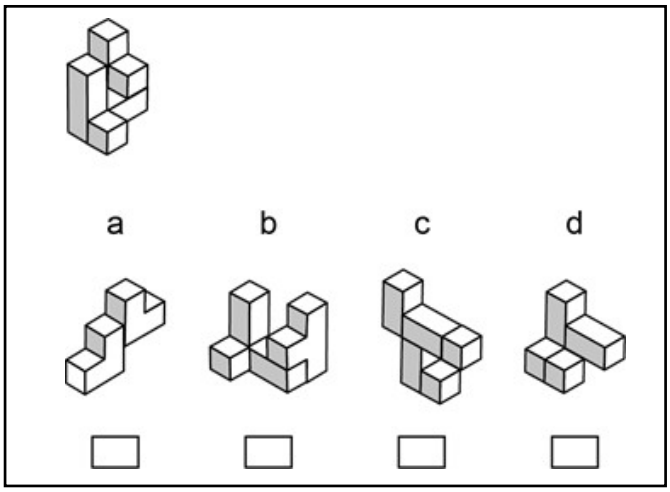
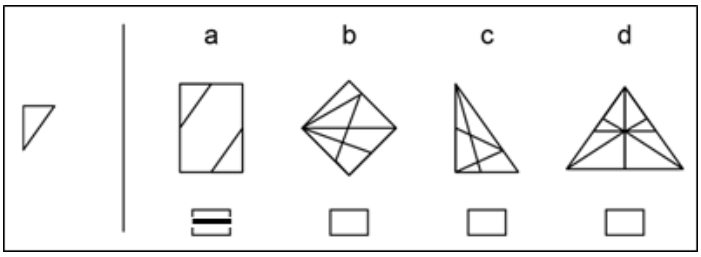


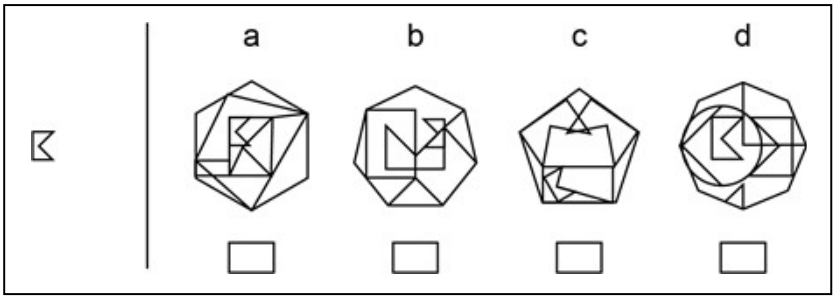
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

Example



1



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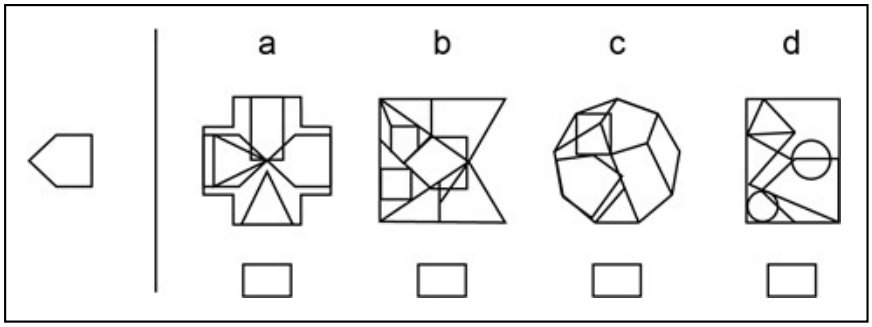

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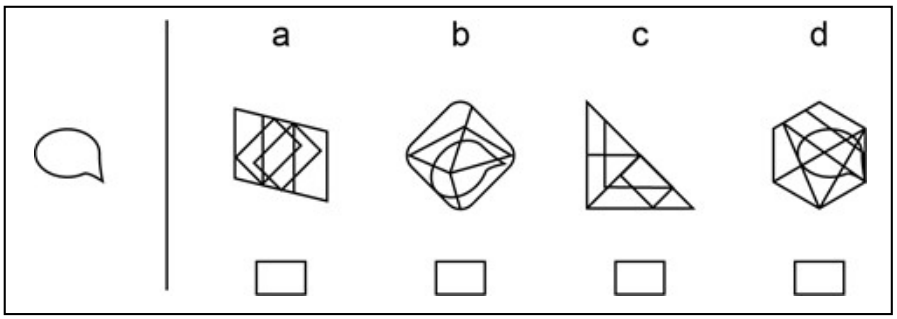
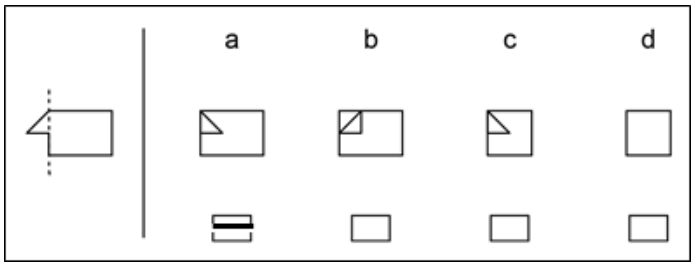


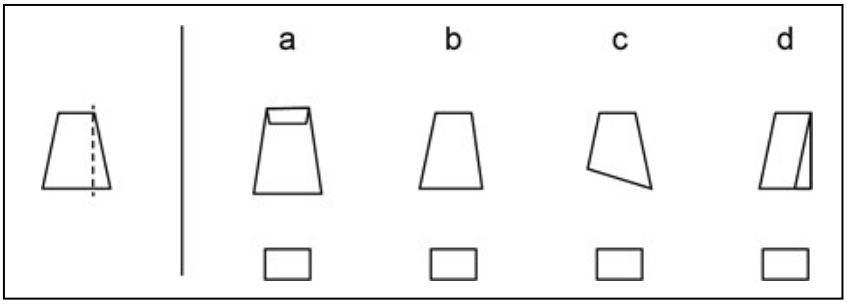
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

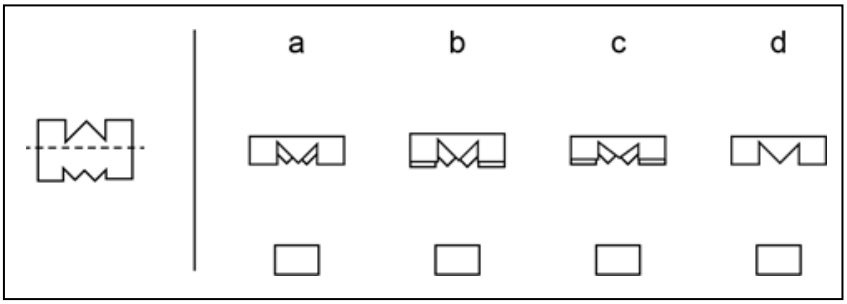
Example



1



2



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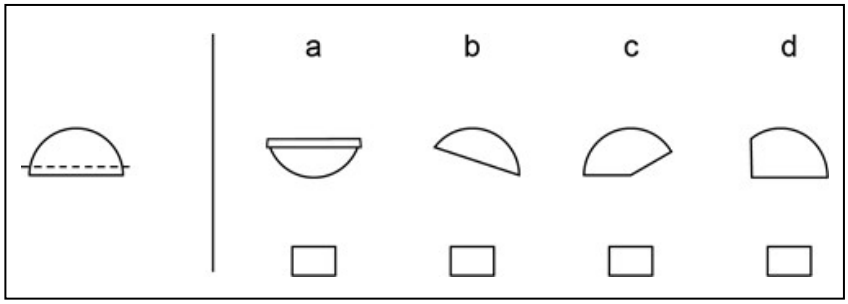
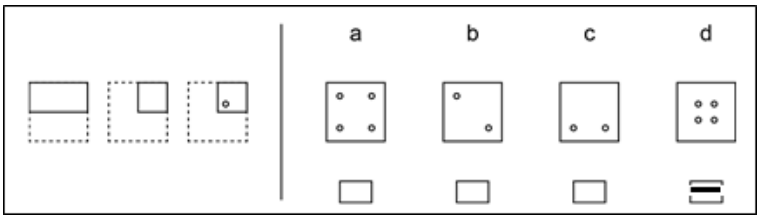


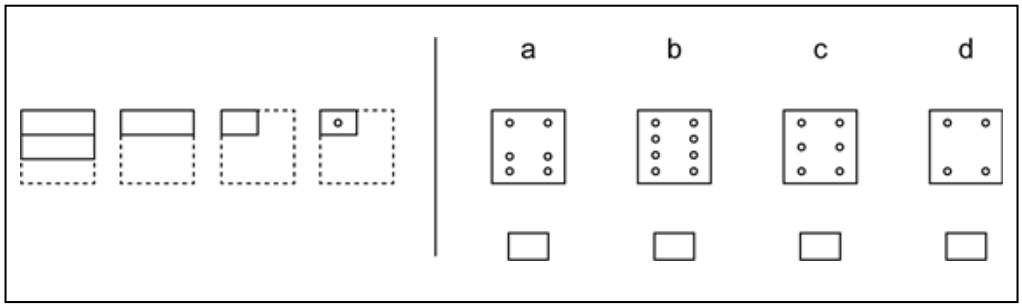
12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

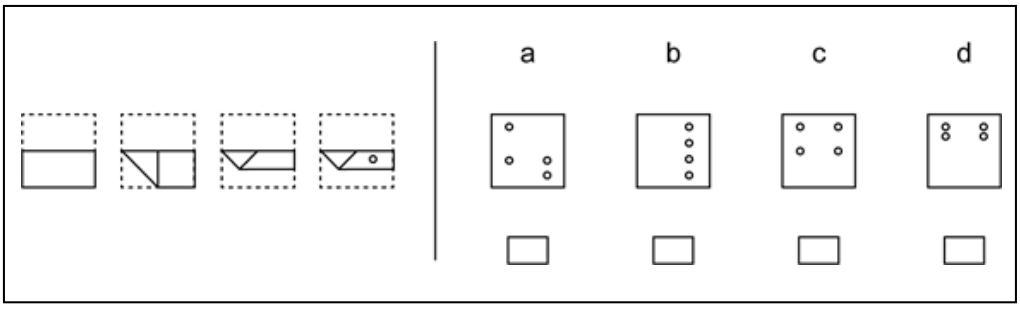
Example



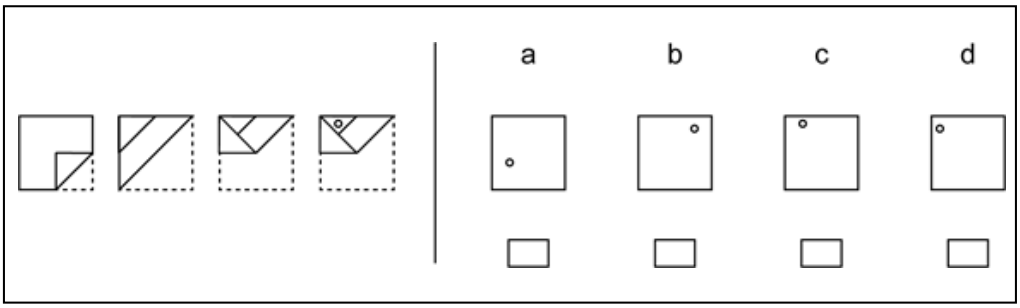
1



2



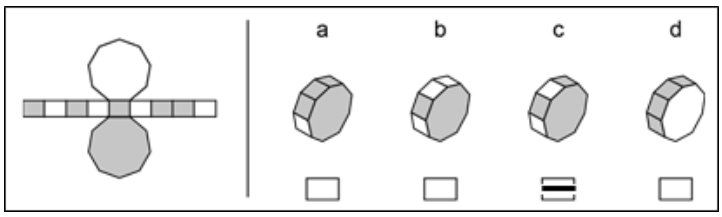
3



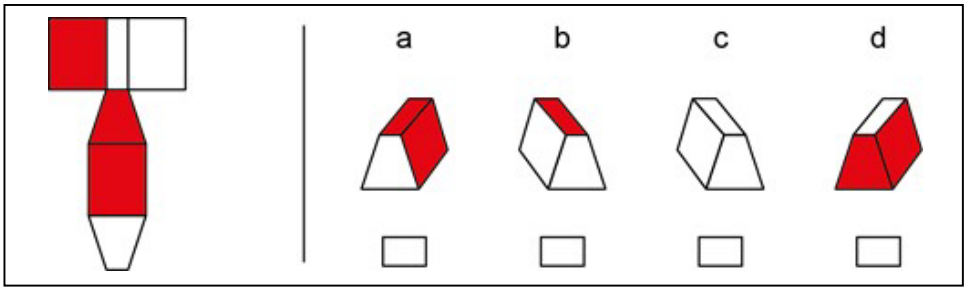
13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

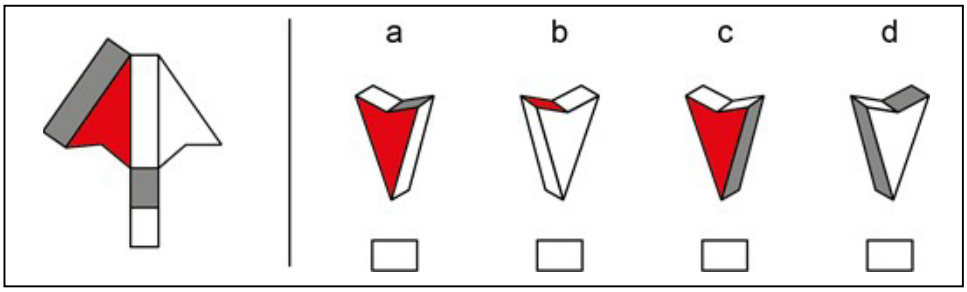
Example



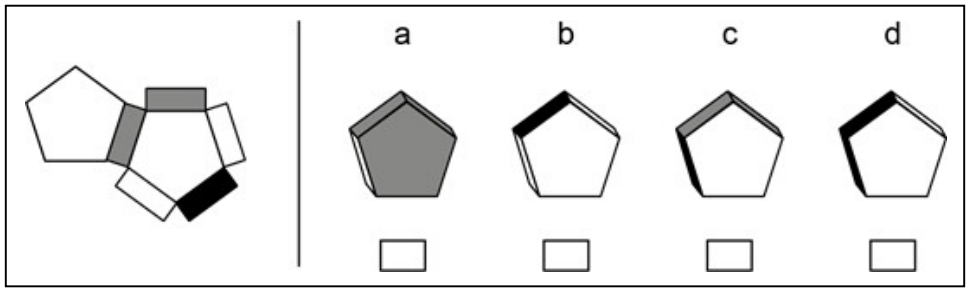
1



2



3



Non-Verbal Reasoning

Like Shapes

1. **A** - (Three changes occur. The original innermost shape becomes the outermost shape. The original outer shape becomes the middle shape. The original black shape becomes the white innermost shape. Therefore the correct answer must be a diamond outer shape, pentagon middle shape and flag inner shape.)
2. **D** - (The first shape has one more side than the second shape. The black circle also moves to the opposite side of the shape and changes colour. Therefore the correct answer must be a two sided shape with a black circle at the bottom.)
3. **C** – (The first and third shapes in the sequence swap positions.)

Odd One Out

1. **A** – (All options have more triangles than circles apart from A.)
2. **B** – (All options share the same pattern of thick, thin, thick concentric rings, apart from B which has thin, thick, thick concentric rings.)
3. **B** – (All options have a quadrilateral as the inner shape apart from B.)

Rotations

1. **A** – (A is a rotation of 90 degrees anti-clockwise.)
2. **C** – (C is a rotation of approximately 135 degrees clockwise.)
3. **B** – (B is a rotation of approximately 135 degrees anti-clockwise.)

Code Breaker

1. **B** - (You can work out that G represents the outer shape being a hexagon as there are two of them. Therefore J stands for the outer shape being a rectangle. You can work out that Q stands for the inner triangle being shaded grey as there are two of them. Therefore O stands for the inner triangle being white. Therefore the answer is JO.)

2. **A** - (You can work out that the second letter represents the shading of the circle as both the second and fourth shapes have K as their second letter and black circles. Therefore L represents the circle being white. You now know that the correct answer must be A or C. The first letter is a bit more difficult to work out. By examining the shapes more closely, the only other difference that the first letter could represent is the relative position of the circle within the pentagon. You will notice that A is at the top, B towards the top right, C towards the bottom right and E towards the top left. You can then deduce that the position that has not been accounted for so far in this clockwise pattern (bottom left) must be D. Therefore the answer is DL.)
3. **B** – (You can work out that F stands for the arrow being horizontal as this is the only similarity between the first and third shapes. Therefore U must represent the arrow being at an angle. You can then deduce that H, D and M represent the respective number of circles at the top as this is the only other difference between the shapes. Therefore the answer is HU.)

Missing Sequence

1. **A** - (Two main changes occur through the sequence. First, larger shape towards the top left is immediately followed by a smaller version of the same shape in the top left corner. Second, the shading of the shapes repeats in the pattern grey, black, white. Therefore the correct answer must be a small white right angled triangle in the top left corner.)
2. **C** – (As you go through the sequence three changes occur: the odd squares contain three black dots and a triangle while the even squares contain four white dots and a square, the triangle in the odd squares alternates between being white and being black, and the triangle in the odd squares also alternates between pointing up and pointing down. Therefore the correct answer must contain three black dots and a white triangle pointing up.)
3. **B** – (The shape rotates by 90 degrees clockwise at each stage of the sequence.)

Missing Square

1. **C** – (Two smaller versions of the shape to the left are mirrored in a curved dividing line.)

2. **B** – (In the square numbers diagonally above and to the right share the same last digit, increasing by 10 each time.)
3. **A** – (The shape rotates by 45 degrees anti-clockwise as you go down each column.)

Most Like

1. **B** – (Both examples show a smaller shape overlapping a larger shape, with a straight line going through both.
2. **D** – (Both examples are a rotation of the same pattern.)
3. **A** – (In both examples there is one more arrow pointing to the left than to the right.)

Nets

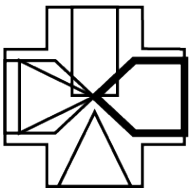
1. **C** - (You can see from the net that the frowning face must be to the left of the winking face and the confused face must be below the winking face.)
2. **B** – (You can see from the net that the star must be underneath the 7 and the equilateral triangle must be to the right of the 7, pointing away from it.)
3. **C** – (You can see from the net that a X must be underneath the & and you can deduce that when folded around the black dot will be to the right of both of them.)

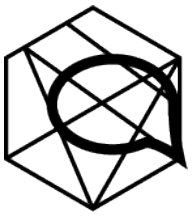
3D Shapes

1. **D** – (Looking at the 3D shape you can see that it must be made up from at least 4 medium sized blocks. You can therefore deduce that the remaining block must be an L shape.)
2. **A** – (When viewed from above the main part of the shape will be a T shape, with an additional square at the top left.)
3. **B** – (B is a rotation of approximately 90 degrees in the y-axis.)

Hidden Shape

1. **A –**

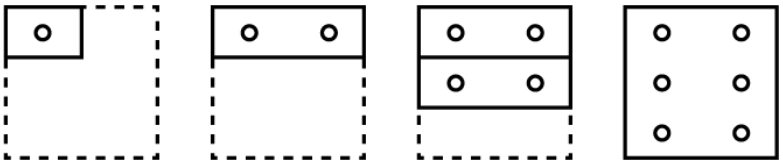
2. **A –** 

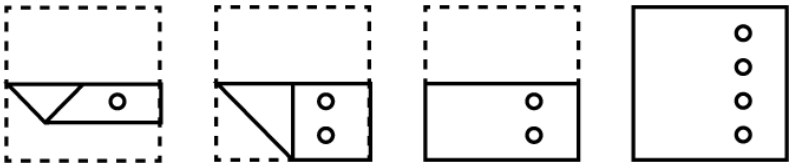
3. **D –** 

Fold Along the Dotted Line

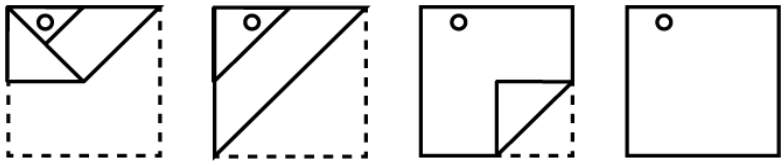
1. **D** – (The right portion will fold over the left portion so that the majority of the left portion is still visible underneath.)
2. **B** – (This question requires close observation. Notice that the fold line is just above half way. This will mean that when the top half folds over the bottom half there will be a small part of the bottom half still visible underneath.)
3. **A** – (The top portion will fold down behind the bottom portion.)

Hole Punch

1. **C –** 

2. **B –** 

3. **C** –



Shaded Nets

1. **C** – (You can see from the net that the top thin rectangular face must be white, therefore A and B cannot be correct. When the net is folded down and the resulting shape flipped upside down so that the shorter side of the trapezium face is at the top, the white trapezium face will have a white rectangular face to the left of it.)
2. **D** – (When the net is folded so that the white arrow face is pointing down, it will have a long grey rectangular face to the left of it. You can see from the net that the small white rectangular face must be next to the longer grey rectangular face.)
3. **B** – (Both pentagons must be white, therefore A cannot be correct. You can see from the shading pattern of the rectangular faces that the black rectangle must have white rectangles either side of it. Therefore B is the correct answer.)